

ALEKSANDRA KARPOVA

Software Engineer | Systems Integration | Industrial Automation

Pavia, Italy [open to relocation]

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Software Engineer with experience developing reliable real-time systems for industrial environments. Skilled in Python, asynchronous processing, OPC UA integration, sensor-data pipelines, Linux-based systems, and modular software architecture for distributed monitoring and control workflows. Experienced in integrating heterogeneous components, designing communication layers, validating system behaviour through simulated environments, and preparing clear technical documentation for engineers and clients. Currently completing a degree in Artificial Intelligence, with a strong interest in maritime automation, remote operations, and autonomous systems.

Work Experience

Software Engineer & ML Engineer– Industrial & Autonomous Systems

OPTECS B.V. | Rotterdam, Netherlands

February 2025 - Present

- Designed Python backend services for real-time processing of 6 industrial sensor tags across 3 data streams, supporting continuous monitoring and control of large-scale mining equipment.
 - Built scalable time-series pipelines for filtering, preprocessing, derived-signal calculation, feature extraction, and control-oriented logic across 8 standalone modules.
 - Developed OPC UA communication layers from scratch on Linux, using asynchronous callbacks, subscriptions, structured node mappings, and tag-driven data models.
 - Created simulated OPC UA environments and structured test scenarios to verify communication flows, edge cases, and module behaviour before deployment.
 - Integrated distributed software modules with clear interfaces, improving maintainability across 10 industrial workflows / services.
 - Developed AI-supported monitoring components for predictive monitoring, anomaly detection, and process optimisation using streaming industrial data.
 - Improved robustness of real-time pipelines by handling missing values, noisy signals, outliers, protocol mismatches, and invalid tag states.
 - Refactored 1,500+ lines of legacy code and increased automated test coverage from ~50% to ~85%, reducing regressions and improving system stability.
 - Produced technical documentation, architecture diagrams, workflow diagrams, and implementation guidance for integration and maintenance.
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Education History

October 2022 - July 2026

BSc in Artificial Intelligence

University of Pavia, University of Milan, University of Milan-Bicocca

- Thesis: Quantum Kernel Classification under QJPEG compression
 - Vice President, AIESEC International (2024–2025)
 - Samsung Innovation Campus – AI & IoT (2023–2024)
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Projects

Satellite Telemetry, Tracking and Control System Personal Project

March 2026 – Present

- Developing a modular spacecraft telemetry and command platform with remote command execution, live telemetry monitoring, system-state transitions, and fault-handling workflows.
- Implementing command/telemetry flows covering command transmission, acknowledgement handling, telemetry updates, and simulated subsystem responses.
- Structuring the platform around 4 modular services/components to support future extensions such as persistent telemetry storage, dashboards, and graphical control interfaces.
- Designing architecture and workflow diagrams documenting command paths, telemetry flow, state transitions, and planned system integrations.
- Building the system as a simulation-first platform to test control logic, communication behaviour, and fault scenarios before adding real hardware interfaces.

Industrial Mill Power Prediction System

Professional Project — Optecs B.V.

February - March 2025

- Developed and evaluated regression models for predicting mill power using industrial process data.
- Compared SGD, Ridge, Lasso, and XGBoost models to identify suitable approaches for predictive monitoring.
- Prepared real-world sensor data for modelling through preprocessing, validation, and feature selection.
- Contributed to the integration of predictive models into a broader industrial monitoring workflow.

OPC UA Adapter for Industrial Systems

Professional Project — Optecs B.V.

June - July 2025

- Designed and developed a reusable OPC UA adapter for communication between heterogeneous industrial systems.
- Implemented asynchronous data exchange, tag mapping, node handling, and validation workflows for real-time industrial applications.
- Supported reliable integration across Linux-based environments through structured error handling and simulated testing scenarios.
- Documented the adapter architecture and communication flows to support integration, maintenance, and future extensions.

Skills

- **Programming and Software Development:** Python, C++, Java, C#, SQL, asynchronous programming, object-oriented programming, REST APIs, FastAPI
- **Industrial Systems and Communications:** OPC UA, asyncua, real-time communications, sensor-data processing, tag-driven systems, SCADA concepts, IIoT concepts, distributed control systems, industrial automation
- **Architecture and Integration:** Modular software architecture, event-driven processing, systems integration, data models, processing models, communication flows, simulated environments, verification workflows
- **Data and Machine Learning:** Scikit-learn, PyTorch, TensorFlow, XGBoost, Pandas, NumPy, time-series preprocessing, feature extraction, regression modelling, anomaly detection
- **Tools and Platforms:** Linux, Docker, Git, CI/CD, Grafana, Prometheus, Parquet
- **Documentation:** Technical reports, architecture diagrams, workflow diagrams, implementation guides, client-facing technical documentation

Certifications & Memberships

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| • Qiskit advocate – IBM Quantum | from February 2026 |
| • AI-SEQ program | September - October 2025 |
| • IBM Machine Learning Professional | May 2024 |

Additional information:

- Strong interest in maritime systems, remote operations, autonomous vehicles, and sustainable industrial automation.
- Former volunteer involved in initiatives connected to the Sustainable Development Goals.
- Open to relocation within Europe.